



Modelling Financial Markets

Chapter 8: Backtesting and Evaluating Risk Forecasts



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ BCBS: Basel Committee on Banking Supervision
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ FDR: False Discovery Rate
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FWER: Family-Wise Error Rate
- ▣ FZ: Fissler–Ziegel (family of scoring functions for VaR and ES)
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ IMA: Internal Models Approach
- ▣ LLM: Large Language Model
- ▣ MCS: Model Confidence Set
- ▣ PIT: Probability Integral Transform

Acronyms used in this material (2/2)

- ☐ POF: **Proportion Of Failures** (test)
- ☐ POT: **Peaks Over Threshold**

Outline

Why Backtest Risk Forecasts?

Backtesting VaR: Coverage and Independence

The Regulator's View: Basel Traffic Light and FRTB

Backtesting Expected Shortfall

Elicitability and Scoring Functions

Comparative Backtesting

Conformal Prediction for Tail Risk

AI for Scientific Discovery

Which Model Would You Certify?

References

A Risk Model Makes a Forecast Every Day

The question of this chapter

- ▣ Every evening a bank reports tomorrow's risk of its trading book
- ▣ The next day the market reveals the actual profit or loss
- ▣ **Backtesting**: comparing past risk forecasts with the outcomes that followed
- ▣ Was the model right **often enough**, and **at the right times**?

What we do in this chapter

- ▣ Test VaR forecasts: coverage, independence, durations
- ▣ Apply the Basel traffic light and the FRTB rules
- ▣ Backtest ES, which is harder, and compare models with scoring functions
- ▣ Try conformal prediction: distribution-free coverage and its limits

Two Risk Measures to Evaluate

- Daily log return $X_t = r_t = \ln(P_t/P_{t-1})$; daily **loss** $L_t = -r_t$ (positive number = money lost)
- **VaR** (Value-at-Risk) at tail probability α : the loss exceeded with probability α
 - ▶ $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$, computed each day from the information $\mathcal{F}_{t-1}$
- **ES** at tail probability α : the average loss in the worst α share of outcomes
 - ▶ $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = E[L_t \mid L_t > \text{VaR}_\alpha]$ (continuous distributions)
- Levels are written as tail probabilities: VaR 1% ($\alpha = 0.01$), VaR 2.5% and ES 2.5% ($\alpha = 0.025$); the target breach rate is α
- Regulation: Basel capital used VaR 1% over 10 days, FRTB uses ES 2.5%; backtests use one-day forecasts. Both measures were built in Chapter 7

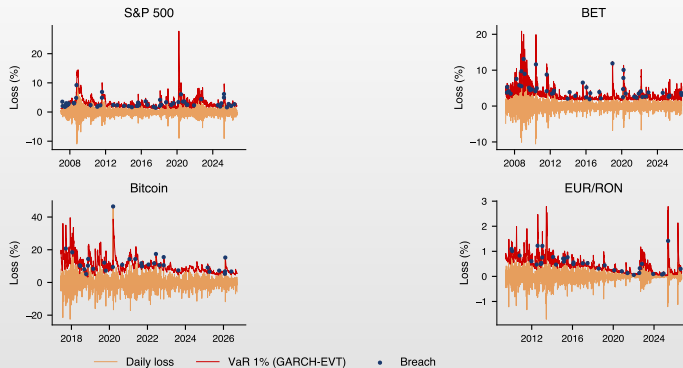
Six Competing Forecasters

- ▣ **HS**: the empirical quantile of order $1 - \alpha$ of the last 1000 losses
- ▣ **Normal**: the Normal distribution with the window mean and standard deviation
- ▣ **Student-t**: a Student-t distribution fitted to the window (location, scale, degrees of freedom ν)
- ▣ **GARCH-t**: $\sigma_t^2 = \omega + a\varepsilon_{t-1}^2 + b\sigma_{t-1}^2$ with standardised Student-t shocks
- ▣ **FHS**: GARCH volatility $\times$ empirical quantiles of the standardised residuals $z_t = \varepsilon_t/\sigma_t$
- ▣ **GARCH-EVT** (McNeil & Frey (2000)): GARCH volatility $\times$ a GPD tail fitted by POT to the largest 10% of standardised losses

The Forecasting Experiment

- ▣ Four series, each on its own trading calendar
 - ▶ S&P 500 and BET (Bucharest Stock Exchange index): closing values
 - ▶ Bitcoin (trades every day) and EUR/RON (BNR reference rate)
- ▣ Rolling window of 1000 observations; each forecast uses only data up to the previous day
- ▣ GARCH, Student-t and EVT parameters re-estimated every 20 days; volatility updated daily
- ▣ Out-of-sample days T (evaluation until 18.09.2026)
 - ▶ S&P 500 4 959 and BET 4 937 (from 2007), EUR/RON 4 352 (from 2009), Bitcoin 3 384 (from 2017)

Four Markets, One Risk Model



Orange: daily losses; red: VaR 1% forecast by GARCH-EVT; blue dots: breaches ($L_t > VaR_t$)

Four Markets: What We See

- ▣ The VaR moves with volatility: it doubles or triples in 2008, March 2020 and the 2022 crypto crash
- ▣ Breach rates of the GARCH-EVT VaR 1%
 - ▶ S&P 500 1.13%, BET 0.99%, Bitcoin 0.95%, EUR/RON 0.78%
- ▣ EUR/RON has a very different scale: a managed exchange rate with long calm periods and rare jumps
- ▣ Question for the chapter: is a rate close to 1% enough to call a model good?

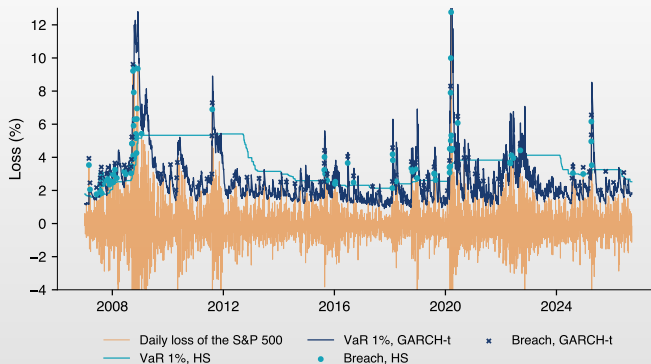
Recap: The Backtesting Problem

- ▣ A risk forecast is a statement about a distribution; we only observe one outcome per day
- ▣ We therefore judge forecasts over many days, with statistical tests
- ▣ Six models, four markets, rolling out-of-sample forecasts: the laboratory of this chapter

The Hit Sequence

- A **breach** (exception, violation) occurs when the loss exceeds the forecast: $I_t = 1\{L_t > \text{VaR}_t^\alpha\}$
- A correct VaR implies two properties ([Christoffersen \(1998\)](#))
 - ▶ **Unconditional coverage:** $P(I_t = 1) = \alpha$, so about αT breaches in T days
 - ▶ **Independence:** yesterday's breach does not change today's breach probability
- Together: I_t are i.i.d. Bernoulli(α) variables
- A model can fail either way: too many breaches, or breaches that come in clusters

S&P 500: Two VaR Forecasts, 2007–2026

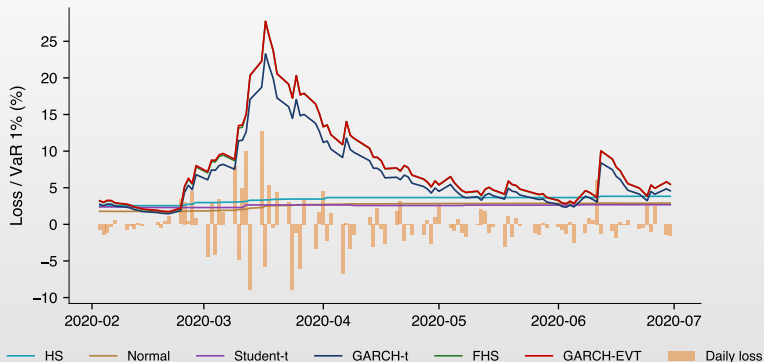


□ VaR 1%: HS in teal, GARCH-t in blue; orange: daily losses; markers: breaches of each model

HS versus GARCH-t: Interpretation

- HS reacts slowly: its VaR is a step function that jumps only when extreme days enter or leave the window
- GARCH-t reacts the next day: the VaR rises as soon as volatility rises
- Almost the same number of breaches: HS 83, GARCH-t 88 in 4 959 days
 - ▶ but HS breaches arrive in bursts (2008, 2020), GARCH-t breaches are spread over time
- Counting breaches is therefore not enough: we must also test **when** they occur

March 2020: The Stress Test Nobody Planned



□ S&P 500, February–June 2020: daily losses (bars) and the VaR 1% of the six models

March 2020: Interpretation

- Largest loss 12.77% on 16.03.2020; in 104 days a correct VaR 1% expects 1.04 breaches
- Breaches in these 104 days
 - ▶ Unconditional models: HS 12, Normal 17, Student-t 17
 - ▶ Conditional models: GARCH-t 3, FHS 2, GARCH-EVT 2
- A fat-tailed distribution (Student-t) does not help if the scale does not move with volatility
- Lesson: in a crisis, conditioning on current volatility matters more than the shape of the tail

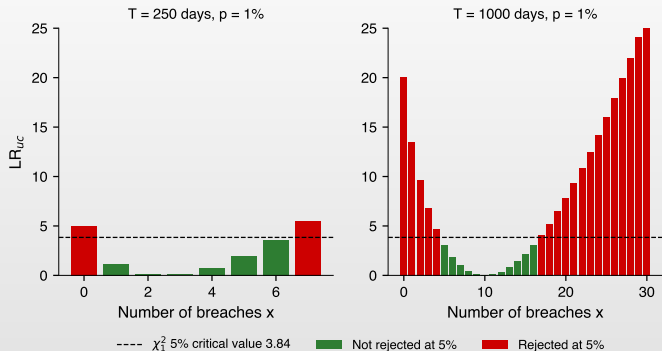
Kupiec (1995): the POF Test

- ▣ Data: $x = \sum_t I_t$ breaches in T days; estimated breach rate $\hat{\pi} = x/T$
- ▣ $H_0: \pi = \alpha$ (correct unconditional coverage), against $H_1: \pi \neq \alpha$
- ▣ Likelihood ratio statistic (Kupiec (1995))
 - ▶ $LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x} \alpha^x}{(1-\hat{\pi})^{T-x} \hat{\pi}^x} \xrightarrow{d} \chi_1^2$
- ▣ Reject at 5% if $LR_{uc} > 3.84$; the test is two-sided: too few breaches also count as failure
- ▣ Exact alternative for small T : the Binomial(T, α) distribution of x

Worked Example: GARCH-t in 2008

- ▣ S&P 500, VaR 1% of GARCH-t, calendar year 2008: $T = 253$ days, $x = 8$ breaches, expected $0.01 \cdot 253 = 2.53$
- ▣ Step 1: $\hat{\pi} = 8/253 = 0.0316$
- ▣ Step 2: $\ln L_0 = 245 \ln 0.99 + 8 \ln 0.01 = -39.30$
- ▣ Step 3: $\ln L_1 = 245 \ln(1 - \hat{\pi}) + 8 \ln \hat{\pi} = -35.50$
- ▣ Step 4: $LR_{uc} = -2(\ln L_0 - \ln L_1) = 7.60 > 3.84$, $p = 0.0058$: coverage rejected
- ▣ With 8 exceptions in about 250 days, a supervisor would place the model in the yellow zone (Section 3)

How Many Breaches Are Acceptable?

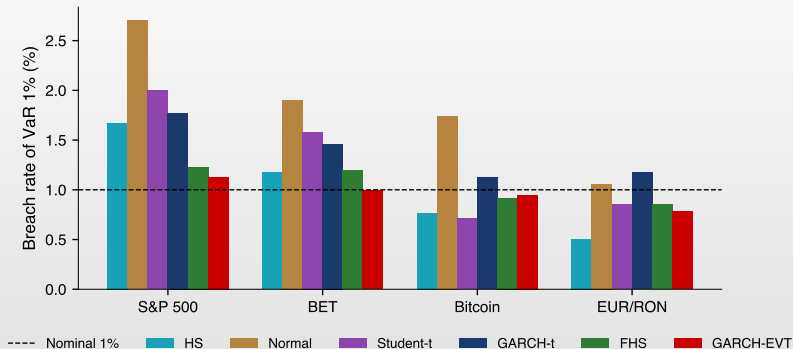


□ LR_{uc} as a function of the number of breaches x , for one year (left) and four years (right) of VaR 1%

The Low Power of Coverage Tests

- ▣ With $T = 250$ the test accepts x from 1 to 6: a model with a true breach rate of 2% (double the target) passes with probability 76%
- ▣ Even $x = 0$ is rejected ($LR_{uc} = -500 \ln 0.99 = 5.03$): a VaR that is never breached is too conservative
- ▣ With $T = 1000$ the acceptance region narrows to 5–16 breaches
- ▣ Tail events are rare by construction: one year of data contains about 2.5 informative days for VaR 1%
- ▣ Consequence: long samples, several levels ($\alpha = 1\%$ and 2.5%) and several tests are needed

Breach Rates Across Models and Markets



□ Share of days with $L_t > \text{VaR}_t$ for the VaR 1%; dashed line: the nominal 1%

Breach Rates: Interpretation

- ▣ The Normal distribution fails everywhere except EUR/RON: S&P 500 2.70%, BET 1.90%, Bitcoin 1.74%
- ▣ GARCH-t is too aggressive on equities (S&P 500 1.77%, BET 1.46%): a symmetric t tail is too thin on the loss side
- ▣ FHS and GARCH-EVT stay closest to 1% in all four markets
- ▣ HS is too conservative on EUR/RON (0.51%): an old crisis in the window keeps its VaR high
- ▣ Kupiec p -values (S&P 500): HS < 0.001 , GARCH-t < 0.001 , FHS 0.116, GARCH-EVT 0.370

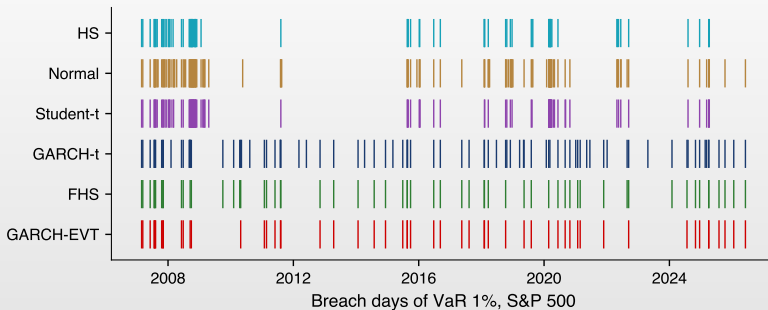
Christoffersen (1998): Testing Independence

- ▣ Model I_t as a first-order Markov chain with $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$
- ▣ Count transitions n_{ij} (from state i yesterday to state j today); $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$
- ▣ H_0 : $\pi_{01} = \pi_{11} = \pi$ (no memory), with $\hat{\pi} = (n_{01} + n_{11}) / \sum n_{ij}$
- ▣ Statistic
 - ▶
$$\text{LR}_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi_1^2$$
- ▣ Conditional coverage: $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind} \xrightarrow{d} \chi_2^2$ tests both properties at once

Worked Example: HS in 2007–2010

- ▣ S&P 500, VaR 1% of HS: $n_{00} = 928$, $n_{01} = 38$, $n_{10} = 38$, $n_{11} = 3$
- ▣ $\hat{\pi}_{01} = 38/(928 + 38) = 0.0393$; $\hat{\pi}_{11} = 3/(38 + 3) = 0.0732$; $\hat{\pi} = 0.0407$
- ▣ After a breach the breach probability almost doubles, yet $LR_{ind} = 0.95$ ($p = 0.33$): not significant
- ▣ Coverage fails badly: $LR_{uc} = 54.24$, so $LR_{cc} = 54.24 + 0.95 = 55.19$
- ▣ Lesson: with only 3 consecutive pairs the Markov test sees almost nothing; clusters spanning several days escape it

When Do Breaches Happen?



- Each tick is one breach of the VaR 1% on the S&P 500

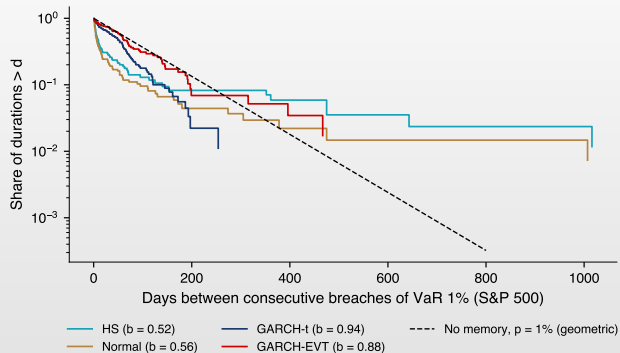
Breach Clustering: Interpretation

- ▣ HS, Normal and Student-t: dense bursts in 2008, 2015–2016, 2018, 2020 and 2022, and long silent periods
- ▣ GARCH-t, FHS and GARCH-EVT: breaches spread almost uniformly over 20 years
- ▣ Clusters mean that losses come **exactly when capital is most needed**
- ▣ Independence p -values (S&P 500): HS <0.001 , Normal <0.001 , GARCH-t 0.095, FHS 0.046

Duration-Based Tests

- Duration D_i : number of days between breach $i - 1$ and breach i
- Under a correct VaR, breaches have no memory: D_i is geometric with mean $1/\alpha$ (100 days for $\alpha = 1\%$)
- Christoffersen & Pelletier (2004): fit a Weibull distribution to the durations
 - ▶ $f(D; a, b) = a^b b D^{b-1} e^{-(aD)^b}$; $b = 1$ is the exponential (memoryless) case
 - ▶ $b < 1$: short durations are too frequent, i.e. breaches cluster
- H_0 : $b = 1$, tested with a likelihood ratio (χ^2_1); the first and last durations are treated as censored
- Uses the whole spacing pattern, not only consecutive days: more power against clustering

How Long Between Breaches?



□ Share of durations longer than d (log scale); dashed: no memory with $\alpha = 1\%$

Durations: Interpretation

- ▣ HS: $b = 0.52$, median duration 7 days, 44% of durations are at most 5 days
- ▣ Normal: $b = 0.56$; GARCH-t: $b = 0.94$ (median 48 days); GARCH-EVT: $b = 0.88$
- ▣ Duration test p -values: HS < 0.001 , GARCH-t 0.479, GARCH-EVT 0.240
- ▣ The duration test rejects HS decisively, where the Markov test for 2007–2010 did not
- ▣ Long tails of HS durations: years without breaches after a crisis leaves a high VaR in the window

All VaR Tests at a Glance

	Kupiec POF				Christoffersen CC				Duration (Weibull)			
HS	<0.001	0.23	0.16	<0.001	<0.001	0.05	0.30	<0.001	<0.001	<0.001	0.25	<0.001
Normal	<0.001	<0.001	<0.001	0.71	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Student-t	<0.001	<0.001	0.07	0.31	<0.001	<0.001	0.08	<0.001	<0.001	<0.001	<0.001	<0.001
GARCH-t	<0.001	0.002	0.48	0.27	<0.001	0.01	0.58	0.48	0.48	0.25	0.96	0.95
FHS	0.12	0.18	0.62	0.31	0.04	0.39	0.51	0.43	0.51	0.12	0.38	0.90
GARCH-EVT	0.37	0.96	0.75	0.13	0.06	0.81	0.57	0.25	0.24	0.20	0.51	0.42
	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON

□ p -values of Kupiec, conditional coverage and duration tests, VaR 1%; red: rejection, green: no rejection

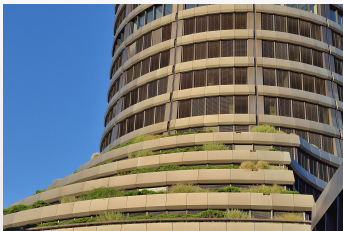
VaR Tests: Interpretation

- Unconditional models (HS, Normal, Student-t) fail the duration test in every market, with one exception: HS on Bitcoin ($p = 0.246$)
- GARCH-EVT is never rejected at 5% by the coverage and duration tests on BET, Bitcoin and EUR/RON
- Bitcoin is the easiest market to pass, probably because every 1000-day window already contains extreme moves
- Several tests on the same breaches are not independent: read the pattern, not a single p -value

Recap: Backtesting VaR

- Kupiec: right number of breaches; Christoffersen: no first-order memory; Christoffersen–Pelletier: no memory in durations
- Coverage tests have low power in one year; durations detect the clustering that coverage misses
- Empirically: conditional models (GARCH filter) beat unconditional ones; the tail model (FHS, EVT) fixes the remaining excess

The Basel Committee and Backtesting



BIS tower, Basel: seat of the BCBS secretariat

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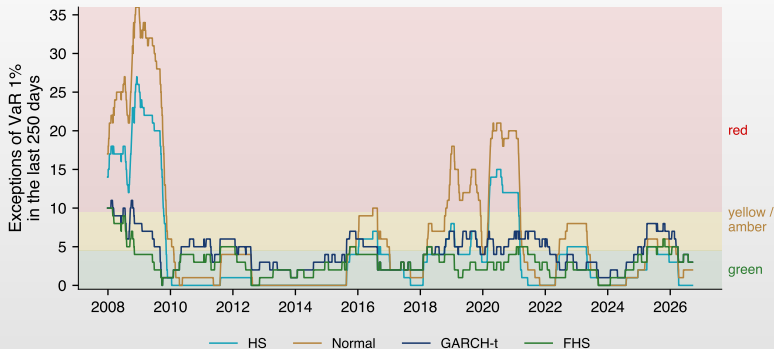
- Banks using internal models must backtest their daily VaR 1% against actual P&L
- BCBS (1996) introduced the **traffic light**: exceptions in the last 250 days determine a capital add-on
- BCBS (2019) (FRTB) keeps the idea, but capital is now based on ES 2.5%
- The regulator wants a rule that is simple, automatic and hard to game

The Traffic Light: Where Do the Zones Come From?

Exceptions	$P(X \leq x)$	Zone
0	8.11%	green
2	54.32%	green
4	89.22%	green
5	95.88%	yellow
7	99.60%	yellow
9	99.97%	yellow
10	99.99%	red

- ▣ $X \sim \text{Binomial}(250, 0.01)$: exceptions of a correct VaR 1% in 250 days
- ▣ Yellow starts where $P(X \leq x) \geq 95\%$, red where $P(X \leq x) \geq 99.99\%$ (BCBS (1996))
- ▣ 1996 rule: the capital multiplier 3 rises by 0.40–0.85 in the yellow zone and by 1 in the red zone
- ▣ Only the **number** of exceptions counts, not their size or timing

The Traffic Light Through Time



□ Exceptions of the VaR 1% in the last 250 days, S&P 500; shaded: green, yellow and red zones

Traffic Light: Interpretation

- ▣ Share of days in the red zone: Normal 22.3%, HS 14.7%, GARCH-t 2.0%, FHS 0.8%
- ▣ Peak: HS 27 and Normal 37 exceptions in the 250 days ending 01.12.2008
- ▣ GARCH-t spends 46.6% of the time in the yellow zone: too many exceptions, but never many at once
- ▣ The rule punishes models **after** a crisis, when capital is already strained: backtesting is backward-looking

FRTB: What Changed in 2019

- ▣ Capital is computed from ES 2.5%, which sees the size of tail losses (BCBS (2019))
- ▣ Backtesting is still done on VaR, at two levels
 - ▶ Bank-wide: VaR 1%, 250 days; multiplier 1.50 (green, 0–4), 1.70–1.92 (amber, 5–9), 2.00 (red, 10 or more)
 - ▶ Trading desk: more than 12 exceptions of VaR 1% or 30 of VaR 2.5% in 12 months sends the desk to the standardised approach
- ▣ Why backtest VaR if capital uses ES? ES alone is not elicitable (Section 5), and VaR tests are simple and robust
- ▣ IMA approval also requires a profit-and-loss attribution test (not covered here)

FRTB Desk Check on the Last 250 Days

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	0 / 2	4 / 7	3 / 4	1 / 5
Normal	2 / 2	6 / 9	3 / 5	1 / 4
Student-t	0 / 2	4 / 9	1 / 4	5 / 8
GARCH-t	3 / 6	4 / 9	3 / 5	4 / 5
FHS	3 / 4	4 / 6	4 / 5	1 / 4
GARCH-EVT	3 / 3	2 / 7	4 / 5	1 / 4

- Cells: exceptions of VaR 1% / of VaR 2.5% in the last 250 days of each series (ending 18.09.2026)
- Limits: 12 for VaR 1%, 30 for VaR 2.5%; every model passes comfortably
- The last year was calm: a green light today says little about the next crisis

Recap: The Regulator's View

- Traffic light = a Binomial test with fixed thresholds, calibrated for 250 days of VaR 1%
- FRTB: capital on ES 2.5%, backtesting on VaR 1% and VaR 2.5%, both bank-wide and per desk
- Unconditional models would have spent long periods in the red zone after 2008 and 2020

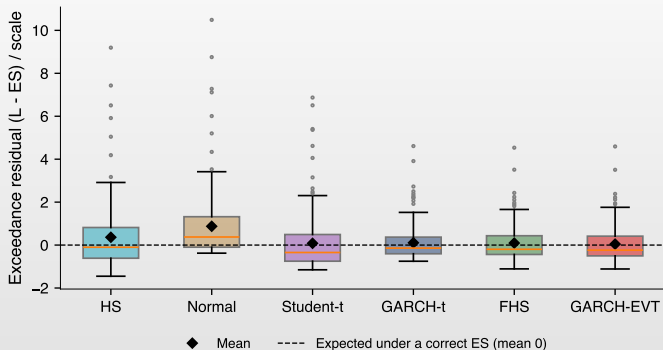
Why ES Is Harder to Backtest

- ▣ A VaR forecast is checked with a yes/no event per day; ES is an **average** over the tail beyond VaR
- ▣ Only the breach days carry information: about αT days (124 for the S&P 500 with $\alpha = 2.5\%$)
- ▣ In 250 days there are $250 \cdot 0.025 = 6.25$ tail observations; the precision of the tail mean scales like $1/\sqrt{\alpha T} \approx 0.40$
- ▣ [Pele & Mazurencu-Marinescu-Pele \(2026\)](#): roughly one in five pairwise ES comparisons in global markets is precision-fragile
- ▣ Three families of tests: residual tests, Acerbi–Szekely statistics, cumulative violations

McNeil and Frey (2000): Exceedance Residuals

- ▣ On each breach day of the VaR 2.5% compute $e_t = (L_t - \text{ES}_t)/\sigma_t$, with σ_t the forecast volatility
- ▣ If ES is correct, $E[L_t - \text{ES}_t \mid L_t > \text{VaR}_t] = 0$, so the residuals have mean zero
- ▣ $H_0: E[e_t] = 0$ against $H_1: E[e_t] > 0$ (ES underestimated); one-sided t -statistic
- ▣ Few and fat-tailed residuals: [McNeil & Frey \(2000\)](#) obtain the p -value with a bootstrap of the centred residuals
- ▣ Scaling by σ_t makes residuals comparable between calm and turbulent days

Exceedance Residuals on the S&P 500



□ Distribution of $(L_t - ES_t) / \sigma_t$ on breach days of the VaR 2.5%; diamonds: means

Exceedance Residuals: Interpretation

- ▣ Means: Normal 0.875, HS 0.363, GARCH-t 0.106, FHS 0.093, GARCH-EVT 0.045
- ▣ Bootstrap p -values: Normal <0.001 , HS <0.001 , GARCH-t 0.023, FHS 0.096, GARCH-EVT 0.246
- ▣ The Normal distribution underestimates the tail mean by almost one volatility unit
- ▣ The outliers (up to 10 volatility units) are the days that decide capital adequacy

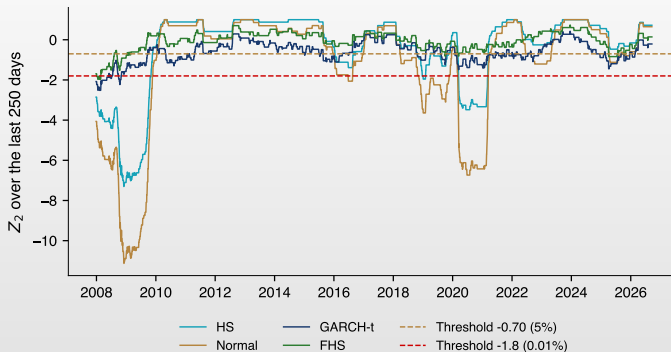
Acerbi and Szekely (2014): Testing ES Directly

- ▣ Breach indicator at the ES level: $I_t = 1\{L_t > \text{VaR}_t^\alpha\}$, $N_T = \sum_t I_t$, $\alpha = 0.025$
- ▣ $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{\text{ES}_t}$: is the average tail loss as large as forecast? (conditional on breaches)
- ▣ $Z_2 = 1 - \frac{1}{T_\alpha} \sum_t \frac{L_t I_t}{\text{ES}_t}$: tests frequency and size together
- ▣ Under H_0 (correct forecast distribution) $E[Z_2] = 0$; underestimated risk gives $Z_2 < 0$ (Acerbi & Szekely (2014))
- ▣ For $T = 250$ and many distributions, the 5% and 0.01% critical values are close to -0.70 and -1.8 : a traffic light for ES

Getting p -Values by Simulation

- The null distribution of Z_2 depends on the forecast distributions F_t , which the model supplies
- Algorithm (repeat $M = 2000$ times)
 - ▶ For each day draw $U_t \sim \text{Uniform}(0,1)$ and set $L_t^* = F_t^{-1}(U_t)$
 - ▶ Compute Z_2^* from the simulated losses with the same VaR and ES forecasts
- p -value = share of simulated Z_2^* at or below the observed Z_2
- Only tail draws matter: for HS and FHS we store the 25 largest window losses of each day
- S&P 500, full sample: $Z_2 = -0.618$ for GARCH-t ($p < 0.001$) versus -0.091 for FHS ($p = 0.172$); critical value about -0.152

Z_2 Over Rolling 250-Day Windows



□ Z_2 of the ES 2.5% forecasts over the last 250 days, S&P 500, with the fixed thresholds -0.70 and -1.8

Z_2 Traffic Light: Interpretation

- Minimum Z_2 : Normal -11.13 and HS -7.30 (window ending 01.12.2008); GARCH-t -2.52 ; FHS -1.97
- Share of windows below -1.8 (red): Normal 21.9%, HS 15.8%, GARCH-t 3.0%, FHS 0.7%
- GARCH-t sits below -0.70 in 39.6% of windows: frequent small underestimation of the tail
- Z_2 punishes the size of the misses, not only their number: HS falls to -7.30 while its exception count peaked at 27

Du and Escanciano (2017): Cumulative Violations

- ▣ PIT: $u_t = F_t(L_t)$, uniform on $(0, 1)$ if the forecast distribution is correct
- ▣ Cumulative violation: $H_t = \frac{1}{\alpha} (u_t - (1 - \alpha)) 1\{u_t > 1 - \alpha\}$: how deep into the tail the loss went
- ▣ Under H_0 : $E[H_t] = \alpha/2$ and $\text{Var}(H_t) = \alpha(1/3 - \alpha/4)$
- ▣ Unconditional test: $U = \sqrt{T} (\bar{H} - \alpha/2) / \sqrt{\alpha(1/3 - \alpha/4)} \xrightarrow{d} N(0, 1)$
- ▣ Conditional test: Box–Pierce statistic on the first 5 autocorrelations of H_t , χ^2_5 (Du & Escanciano (2017))

All ES Tests at a Glance

	McNeil-Frey				Acerbi-Szekely Z2				Du-Escanciano U			
HS	<0.001	0.48	0.70	0.46	<0.001	0.13	0.86	1.00	<0.001	0.36	0.14	<0.001
Normal	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	0.97	<0.001	<0.001	<0.001	0.25
Student-t	0.17	0.01	1.00	0.47	0.002	0.003	0.62	0.92	<0.001	<0.001	0.93	0.05
GARCH-t	0.02	0.04	0.50	0.41	<0.001	<0.001	0.05	0.63	<0.001	<0.001	0.13	0.86
FHS	0.10	0.63	0.60	0.75	0.17	0.19	0.35	0.97	0.09	0.46	0.95	0.09
GARCH-EVT	0.25	0.57	0.29	0.69	0.09	0.21	0.58	0.98	0.24	0.54	0.70	0.11
	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON

□ p -values for ES 2.5%: McNeil-Frey (bootstrap), Z_2 (simulation), Du-Escanciano U

ES Tests: Interpretation

- The Normal distribution is rejected by all three tests on S&P 500, BET and Bitcoin
- FHS and GARCH-EVT pass all ES tests at 5% on all four markets
- GARCH-t fails Z_2 on equities ($p < 0.001$ on S&P 500, < 0.001 on BET): the t tail is too thin
- EUR/RON: most models over-predict ES ($Z_2 > 0$), a consequence of the managed exchange rate
- The tests agree on the ranking but not on marginal cases: Student-t on the S&P 500 has McNeil–Frey $p = 0.173$ but Z_2 $p = 0.002$

Recap: Backtesting ES

- ES tests use only the tail days: expect low power over one year
- McNeil–Frey tests the mean residual, Z_2 tests frequency and size, Du–Escanciano tests the depth of violations
- p -values need the full forecast distribution (simulation or PIT), not only VaR and ES numbers

Testing Is Not Ranking

- ▣ A backtest asks: is this model **acceptable**? Several models can pass, or all can fail
- ▣ Comparative backtesting asks: which model is **better**? (Nolde & Ziegel (2017))
- ▣ Ranking needs a **scoring function** $S(x, L)$: the loss of reporting x when L occurs
- ▣ Compare the average scores $\bar{S} = \frac{1}{T} \sum_t S(x_t, L_t)$: lower is better
- ▣ But which scoring function? A bad one can reward a wrong forecast

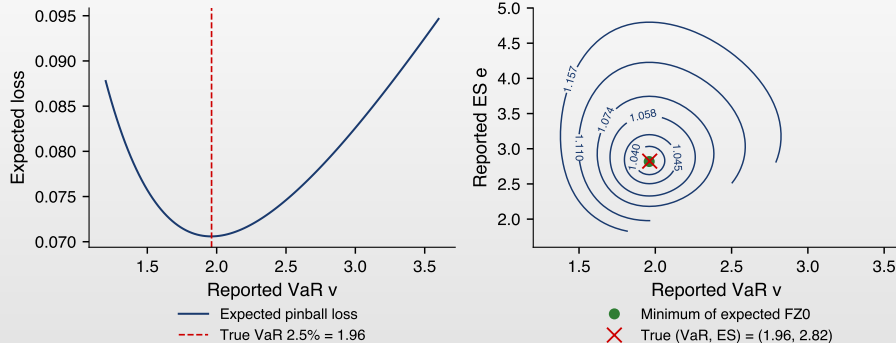
Elicitability

- ▣ A statistic $\theta(F)$ is **elicitable** if some scoring function is minimised in expectation exactly by the true value (Gneiting (2011))
- ▣ $\theta(F) = \arg \min_x E_F [S(x, L)]$
- ▣ Mean: squared error $S(x, L) = (x - L)^2$
- ▣ Quantile, i.e. VaR: pinball loss $S(v, L) = (1\{L > v\} - \alpha)(L - v)$
- ▣ ES is **not** elicitable on its own: no scoring function of e alone is minimised by the true ES (Gneiting (2011))

VaR and ES Together: Fissler and Ziegel (2016)

- The pair (VaR, ES) is jointly elicitable: [Fissler & Ziegel \(2016\)](#) characterise the whole family of consistent scoring functions
- FZ0 loss ([Patton, Ziegel & Chen \(2019\)](#)), with losses positive and $0 < v \leq e$
 - ▶
$$\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$$
- First term: penalty for breaches, scaled by ES; last terms: penalty for reporting a large ES
- Differences of FZ0 losses do not depend on the units of the losses, so they can be compared across assets
- Lower average FZ0 = better joint VaR and ES forecast

Elicitability in Pictures



□ Student-t losses with $\nu = 4$ and unit variance, $\alpha = 2.5\%$; expected losses computed on 2 million draws

Elicitability: Interpretation

- Left: the expected pinball loss is lowest at the true VaR 2.5%, 1.96
- Right: the expected FZ0 loss is lowest at (1.96, 2.82), i.e. at the true pair (VaR, ES) up to the grid resolution
- A forecaster who knows the truth has no incentive to report anything else: the score is **consistent**
- Consistency is what makes ranking by average score meaningful

Worked Example: FZ0 on Two Days

- ▣ Forecast (in %): VaR $v = 3.0$, ES $e = 3.9$, $\alpha = 0.025$; constant part $v/e + \ln e - 1 = 0.769 + 1.361 - 1 = 1.130$
- ▣ Day 1, loss $L = 1.0 < v$: no breach, FZ0 = 1.130
- ▣ Day 2, loss $L = 4.5 > v$: breach penalty $(4.5 - 3.0)/(0.025 \cdot 3.9) = 15.385$, so FZ0 = 16.515
- ▣ One breach costs as much as many quiet days: the score is dominated by the tail
- ▣ A larger ES lowers the breach penalty but raises $\ln e$: the optimum balances the two

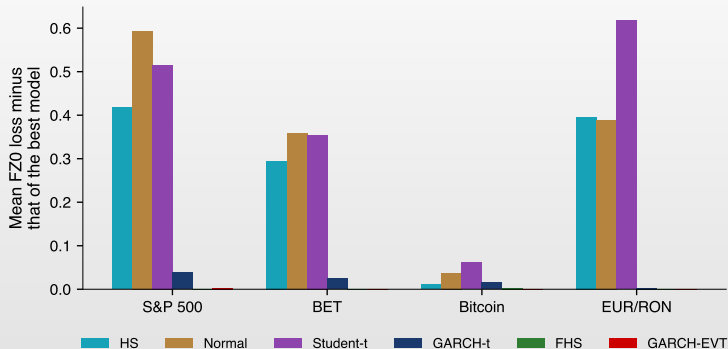
Recap: Elicitability

- ▣ Elicitable = there is a scoring function that rewards honesty
- ▣ VaR: pinball loss; ES alone: impossible; (VaR, ES) jointly: FZ family, e.g. FZ0
- ▣ Elicitability matters for **ranking** models; ES can still be **tested** (Section 4)

The Diebold–Mariano Test on FZ0 Losses

- Loss differential of models A and B : $d_t = \text{FZ0}_{A,t} - \text{FZ0}_{B,t}$
- H_0 : $E[d_t] = 0$ (equal predictive accuracy), [Diebold & Mariano \(1995\)](#)
- $$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d_t)/T}} \xrightarrow{d} N(0, 1)$$
- $\widehat{\text{LRV}}$: long-run variance with Newey–West weights ([Newey & West \(1987\)](#)), 9 lags; tail losses cluster, so d_t is autocorrelated
- $\text{DM} > 0$: model A has the larger average loss (worse); $|\text{DM}| > 1.96$: significant at 5%

Average FZ0 Loss by Model

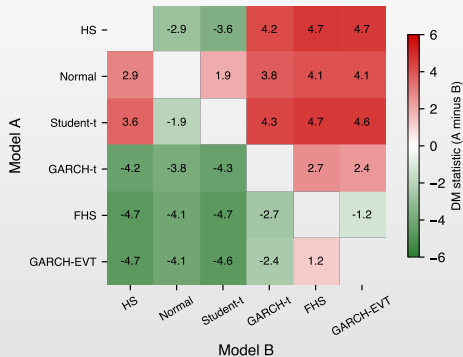


□ Mean FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts minus that of the best model in each market

FZ0 Ranking: Interpretation

- ▣ Best model: FHS on the S&P 500; GARCH-EVT on BET, Bitcoin and EUR/RON
- ▣ Unconditional models lose by 0.3–0.6 FZ0 units on S&P 500, BET and EUR/RON
- ▣ On Bitcoin all models are close (largest gap 0.062): the data cannot separate them
- ▣ GARCH-t is behind FHS on the S&P 500 by 0.039: small, but is it significant?

Pairwise Diebold–Mariano Tests, S&P 500



□ DM statistic of model *A* (row) minus model *B* (column) on FZ0 losses; red: *A* worse, green: *A* better

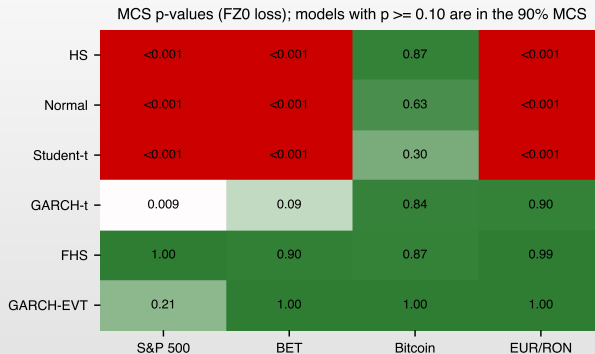
Diebold–Mariano: Interpretation

- ▣ HS versus FHS: $DM = 4.70$; GARCH-t versus FHS: $DM = 2.65$: both significantly worse
- ▣ GARCH-EVT versus FHS: $DM = 1.24$: statistically indistinguishable
- ▣ HS beats the Normal distribution ($DM = 2.88$ for Normal minus HS): a slow empirical quantile is still better than a thin tail
- ▣ Many pairwise tests raise a multiple-testing problem: which models survive **jointly**?

The Model Confidence Set

- Hansen, Lunde & Nason (2011): the smallest set of models that contains the best one with probability $1 - \alpha_{MCS}$
- Algorithm
 - ▶ Test H_0 : all models in the current set have equal expected loss
 - ▶ If rejected, eliminate the model with the largest standardised average loss and repeat
 - ▶ Critical values from a block bootstrap (blocks of 10 days, 1000 replications)
- MCS p -value of a model: the largest test p -value up to its elimination; keep models with $p \geq 0.10$
- A large MCS is an honest answer: the data do not separate the models

Which Models Survive?



□ MCS p -values for the FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts, per market

MCS: Interpretation

- ▣ S&P 500: only FHS and GARCH-EVT (MCS p 0.21) remain; GARCH-t is excluded (p 0.01)
- ▣ EUR/RON: the three conditional models remain; BET: FHS and GARCH-EVT remain, GARCH-t just misses (p 0.09)
- ▣ Unconditional models are excluded everywhere except Bitcoin
- ▣ Bitcoin: all six models remain: the data cannot separate them (compare [Pele & Mazurencu-Marinescu-Pele \(2026\)](#))
- ▣ Conclusion across tests: FHS and GARCH-EVT are the models to beat

Recap: Comparative Backtesting

- ▣ Rank models with a consistent score (FZ0 for VaR and ES); compare them with DM tests using HAC variances
- ▣ The MCS handles many models at once and says how many are statistically equivalent
- ▣ A model can pass all backtests and still be dominated by a better one

A Distribution-Free Promise



Emmanuel Candès, co-author of adaptive conformal inference

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Wikimedia Commons

- All models so far assume a distribution (Normal, t , GPD) or a stable history, as HS does
- **Conformal prediction** turns any forecast into a quantile with a coverage guarantee (Lei et al. (2018), Angelopoulos & Bates (2021))
- The only assumption: the data are **exchangeable**, i.e. their joint distribution does not change under reordering
- Financial returns have volatility clustering: exchangeability is exactly what fails

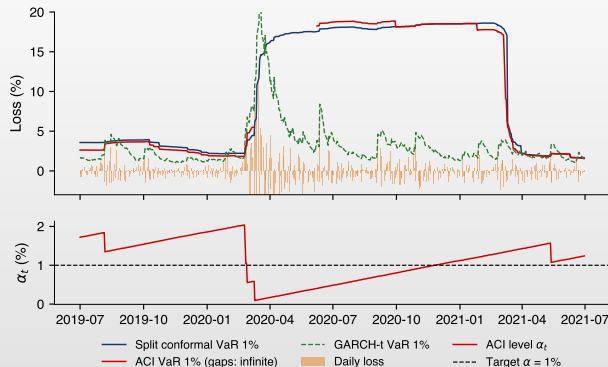
Split Conformal Quantiles

- Nonconformity score: $s_t = L_t / \hat{\sigma}_t$, with $\hat{\sigma}_t$ the standard deviation of the previous 250 returns
- Calibration set: the last $n = 250$ scores; sort them $s_{(1)} \leq \dots \leq s_{(n)}$
- Conformal VaR: $\text{VaR}_t = \hat{\sigma}_t s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$
- Guarantee if the $n+1$ scores are exchangeable: $P(L_t \leq \text{VaR}_t) \geq 1-\alpha$ (and $\leq 1-\alpha + \frac{1}{n+1}$ without ties)
- Example: $n = 250$, $\alpha = 0.01$: $k = \lceil 251 \cdot 0.99 \rceil = 249$, the second largest score; breach probability between 0.60% and 1%

Adaptive Conformal Inference

- Gibbs & Candès (2021): update the working level after each day, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, $\text{err}_t = 1\{L_t > \text{VaR}_t\}$
- After a breach α_t falls (VaR rises); after calm days it slowly rises again; we use $\gamma = 0.005$
- Guarantee without any assumption on the data: $\left| \frac{1}{T} \sum_t \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$
- The guarantee is on the long-run **average**, not on each day or each crisis
- If $\alpha_t < 1/(n+1)$, the required quantile does not exist: the VaR becomes infinite

Conformal VaR Through COVID-19

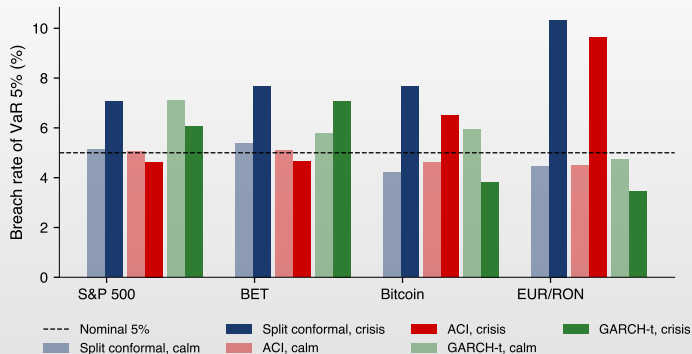


□ S&P 500, July 2019–June 2021: split conformal, ACI and GARCH-t VaR 1% (top) and the ACI level α_t (bottom)

Conformal VaR in a Crisis: Interpretation

- ▣ 20.02–30.04.2020 (50 days, 0.5 breaches expected): split conformal 6 breaches, ACI 4
- ▣ ACI reached its target by setting $\text{VaR} = \infty$ on 37 of these 50 days: coverage without information
- ▣ After the crash, the calibration set contains March 2020: the split conformal VaR stays near 18% for a whole year
- ▣ GARCH-t adapts in days because it models the dependence; conformal methods only react to breaches

Coverage in Calm and Crisis Periods



- Breach rates of the VaR 5%; crisis days: volatility of the previous 20 days above its 90th percentile (known in advance)

Calm Versus Crisis: Interpretation

- Over all days split conformal is close to 5%: S&P 500 5.34%, BET 5.63%, Bitcoin 4.58%, EUR/RON 5.06%
- After turbulent periods it breaches more: S&P 500 7.06%, BET 7.69%, Bitcoin 7.67%, EUR/RON 10.32%
- ACI repairs the equity markets (S&P 500 4.64%, BET 4.66%) but not EUR/RON (9.63%)
- For VaR 1% ACI keeps the average only by an infinite VaR on 13.4% of S&P 500 days
- Marginal coverage holds, conditional coverage does not; a promising route is to conformalise a good conditional model (scores standardised by GARCH volatility)

Recap: Conformal Prediction

- Split conformal: finite-sample coverage under exchangeability, with no distributional model
- ACl: long-run average coverage for any sequence, possibly with infinite VaR
- Both fail conditionally in volatility clusters: backtest them like any other model

An Open Question: Conditional Coverage in Crises

- Distribution-free VaR guarantees are **marginal**: split conformal needs exchangeability, ACI only controls the long-run average
 - ▶ Barber, Candès, Ramdas & Tibshirani (2023): without exchangeability the coverage loss is bounded by distances between the calibration and test distributions, which volatility clustering makes large
 - ▶ Gibbs, Cherian & Candès (2025): exact coverage conditional on a pre-specified class of shifts is attainable; which class captures a financial crisis is open
- Formal question, with a crisis flag $C_t \in \mathcal{F}_{t-1}$ known ex ante
 - ▶ H_0 : $P(L_t > \text{VaR}_t \mid C_t = 1) = \alpha$ against H_1 : $P(L_t > \text{VaR}_t \mid C_t = 1) \neq \alpha$
 - ▶ Falsified by a crisis breach rate outside the Binomial acceptance region, after correcting for dependence between breaches and for multiplicity
- Stakes: capital is needed exactly on the days when marginal guarantees say least

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: map the guarantees, then check every DOI on [Crossref](#) (the title must match)
 - ▶ List peer-reviewed work since 2019 on conformal prediction under serial dependence; for each, give the DOI, the exact guarantee (marginal, conditional or long-run) and its assumptions.
- 2. **Competing hypotheses**, each with a testable implication
 - ▶ Give three rival explanations for why split conformal VaR over-breaches after volatile periods (stale scale, changing tail shape, regime-selection artefact) and what each predicts for breach rates by regime.
- 3. **Data and code**: replicate a known number before changing anything
 - ▶ Write Python that reads data/market/GSPC.INDX.csv, builds the split conformal VaR 5% of this chapter with strictly lagged inputs and prints the crisis breach rate. Target: 7.06%

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, written down before the results are seen
 - ▶ Draft a pre-registered robustness plan: `alpha in {1%, 5%}`, `calibration size in {125, 250, 500}`, `crisis cut-off at the 80th, 90th and 95th percentile`, `two subsamples`, `Holm correction across markets`.
 - ▶ Placebo: random crisis labels with the same frequency must give a crisis breach rate equal to α
- 5. **Critique**: ask the assistant to attack the result
 - ▶ Act as a hostile referee: `list the strongest reasons this crisis effect is an artefact (clustered breaches, regime defined from the same data, overlapping windows) and the test that would settle each`.
- 6. **Report**: every specification tried, effect sizes with intervals, code, and the log of AI use

What the Human Must Check

- **Look-ahead:** the scale $\hat{\sigma}_t$, the GARCH parameters and the crisis flag may use only $\mathcal{F}_{t-1}$; test: recompute with every input lagged one more day
- **Dependent breaches:** clustered hits make Binomial p -values too small; test: Christoffersen and duration tests, or a block bootstrap of the hit sequence
- **Power:** S&P 500 has 471 crisis days, i.e. 4.7 expected breaches of VaR 1%; a true rate of 2% is detected with probability 47%; compute power before testing
- **Multiple testing:** 4 markets $\times$ 2 levels $\times$ 3 methods = 24 tests; control the FWER (Holm) or the FDR (BH)
- **Misstated theorems and invented references:** e.g. “ACI guarantees conditional coverage”; check the theorem in [Gibbs & Candès \(2021\)](#) and every DOI
- **Off-by-one quantiles:** the index $k = \lceil (n+1)(1-\alpha) \rceil$ is not the default of `np.quantile`; test on simulated i.i.d. data

Mini-Case: Does a GARCH Scale Repair Crisis Coverage?

- Replication: the split conformal VaR 5% of this chapter breaches on 7.06% of S&P 500 crisis days (Section 7)
- A typical assistant proposal: score $s_t = (L_t + \mu_t)/\sigma_t$ with the GARCH-t forecasts; H_0 : crisis breach rate = 5%, exact Binomial test, Holm across markets

Market	Crisis days	Split conformal	Holm p	GARCH-scaled	Holm p
S&P 500	471	6.37%	0.169	1.06%	<0.001
BET	469	7.89%	0.023	4.05%	0.794
Bitcoin	314	7.96%	0.053	3.18%	0.464
EUR/RON	411	9.25%	0.001	4.62%	0.821

- Result: coverage restored on BET, Bitcoin and EUR/RON, but the S&P 500 swings to 1.06%: the GARCH scale overshoots after spikes; next hypothesis: the error is in the scale dynamics, not in the tail
- Common sample: 4 709 S&P 500 days after 250 calibration days; code: `Quantlets/Ch_08/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: can a distribution-free VaR 1% and ES 2.5% forecast achieve crisis-conditional coverage in all four markets under a pre-registered test?
- Replicate first: the crisis breach rates of split conformal VaR 5% (S&P 500 7.06%, EUR/RON 10.32%) and the ACI share of infinite VaR 1% (13.4%)
- Deliverables
 - ▶ A pre-registration (hypotheses, tests, corrections, power) dated before the first run
 - ▶ A candidate method, e.g. [Gibbs, Cherian & Candès \(2025\)](#) with the volatility regime as the conditioning class, backtested with Kupiec, Christoffersen, duration and Z_2 tests
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`:
github.com/danpele/MFM/tree/main/project

Scorecard Across Four Markets

- Each cell: breach rate of the VaR 1% in % / Z_2 p -value / MCS p -value (FZ0)

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	1.67 / <0.001 / <0.001	1.17 / 0.134 / <0.001	0.77 / 0.863 / 0.87	0.51 / 1.000 / <0.001
Normal	2.70 / <0.001 / <0.001	1.90 / <0.001 / <0.001	1.74 / 0.001 / 0.63	1.06 / 0.973 / <0.001
Student-t	2.00 / 0.002 / <0.001	1.58 / 0.003 / <0.001	0.71 / 0.625 / 0.30	0.85 / 0.920 / <0.001
GARCH-t	1.77 / <0.001 / 0.01	1.46 / <0.001 / 0.09	1.12 / 0.050 / 0.84	1.17 / 0.630 / 0.90
FHS	1.23 / 0.172 / 1.00	1.20 / 0.188 / 0.90	0.92 / 0.345 / 0.87	0.85 / 0.968 / 0.99
GARCH-EVT	1.13 / 0.090 / 0.21	0.99 / 0.210 / 1.00	0.95 / 0.581 / 1.00	0.78 / 0.976 / 1.00

- Only FHS and GARCH-EVT pass the Z_2 test at 5% and stay in the 90% MCS in all four markets

A Romanian Bank's Trading Book



National Bank of Romania, Bucharest

Photo: Britchi Mirela (2018); CC BY-SA 4.0; Wikimedia

[Commons](#)

- Typical exposures: Romanian equities, via the BET index, and the EUR/RON exchange rate
- BET: HS and the Normal distribution fail the duration tests; GARCH-EVT passes all VaR and ES tests
- EUR/RON: long calm spells and rare jumps; HS breaches only 0.51% of days, and all unconditional models fail the independence test
- Seminar 8, Part C: which model would you certify for this book?

From Our Research Group

- ▣ Pele, Lazar & Dufour (2017): entropy of intraday returns predicts the daily VaR of EUR/JPY
- ▣ Pele & Mazurencu-Marinescu-Pele (2019): entropy-based VaR forecasts for Bitcoin, backtested with Christoffersen's tests
- ▣ Pele, Lazar & Mazurencu-Marinescu-Pele (2019): a tail-entropy measure of ES and its backtesting performance
- ▣ Pele & Mazurencu-Marinescu-Pele (2026): when are two ES forecasts statistically distinguishable?
- ▣ Textbooks: Franke, Härdle & Hafner (2019) (VaR and backtesting chapters) and Härdle, Chen & Overbeck (2017)

Further Study on Quantinar

Advanced course on quantinar.com

 Machine learning in Financial Risk

Code and data

-  All charts and tests: github.com/danpele/MFM/tree/main/Quantlets/Ch_08
-  Series: S&P 500, BET, Bitcoin (repository folder `data/market`); EUR/RON: BNR reference rate

Key Takeaways

- ▣ Test both the number and the timing of breaches: coverage, independence, durations
- ▣ Regulators use simple counts (traffic light), now combined with ES-based capital under FRTB
- ▣ ES can be backtested (Z_2 , residuals, cumulative violations) but not ranked alone: use FZ0 for (VaR, ES)
- ▣ Rank with DM tests and the MCS; accept that some models cannot be separated
- ▣ Conformal guarantees are marginal: they do not protect you in the crisis
- ▣ Empirically, a GARCH filter plus a flexible tail (FHS, EVT) wins in all four markets

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